



Cosmic Pioneers: The Role of Insects in Advancing Space Biology and Sustainable Space Habitats

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Abstract

Insects have served as invaluable models in space biology, revealing how life adapts to microgravity, cosmic radiation and limited resources. Beginning with fruit flies launched aboard V-2 rockets in 1947, research has expanded to study genetics, neurobiology, immunity and reproductive adaptation in *Drosophila melanogaster*. Recent work explores edible insects such as crickets and mealworms as sustainable protein sources for long-duration missions. These studies provide insights into human health, ecological interactions and space agriculture. By tracing the scientific contributions of insects, this review highlights their role as “tiny pioneers” guiding humanity toward safe, sustainable and biologically informed exploration beyond Earth.

Keywords: Insects, *Drosophila melanogaster*, edible insects, space agriculture and immunity

Introduction

When most people imagine space exploration, they think of astronauts in bulky suits, gleaming rockets, or robotic rovers crossing Martian landscapes. Rarely do insects come to mind. Yet since the dawn of the space age, insects have quietly accompanied humanity beyond Earth, serving as living test subjects, biological indicators, and even potential food sources for future explorers. In 1947, fruit flies became the first animals launched into space, strapped aboard a U.S. V-2 rocket to study the effects of cosmic radiation on living organisms. Their survival marked the beginning of a long scientific journey that would see insects playing crucial roles in understanding how life responds to microgravity, radiation and isolation (Valdivia-Silva *et al.*, 2024). Unlike larger animals, insects are small, easy to breed and capable of producing many generations in a short time. These qualities make them particularly valuable for space experiments where resources are limited and results must be observed quickly. Among insects, *Drosophila melanogaster* (Fruit fly) has become a superstar of space biology. Already one of the most studied species in genetics and developmental biology, fruit flies are now helping scientists understand how living systems cope with weightlessness and space radiation (Ma *et al.*, 2015; Ogneva *et al.*, 2016).

But insects’ role in space goes far beyond being passive test subjects. Studies have revealed how they adapt at the cellular level, how their immune systems respond to infection in microgravity, and even how artificial gravity can mitigate space-induced neurological changes (Mhatre *et al.*, 2022; Taylor *et al.*, 2014). In parallel, edible insects such as crickets (*Acheta domesticus*) and mealworms (*Tenebrio molitor*) are being studied as sustainable protein sources for long-duration missions, offering astronauts nutritious food while reducing reliance on resource-intensive cargo from Earth (Guidetti *et al.*, 2025). By tracing these developments,

this article illustrates how insects-creatures often overlooked in daily life-have become vital allies in one of humanity’s greatest adventures. They are not only helping us survive in space but also shaping our vision of how life itself might flourish beyond Earth.

Tiny Astronauts, Giant Leaps: The Journey of Insects Beyond Earth

Early Space Pioneers

The story of insects in space began with fruit flies aboard a V-2 rocket in 1947. This first mission was not about curiosity-it was about survival. Scientists wanted to know how cosmic radiation affected living organisms. Fruit flies, with their short life cycle and known genetic makeup, were ideal candidates. They endured the flight and became the first living animals to return from beyond Earth’s atmosphere, paving the way for mammals and, eventually, humans (Valdivia-Silva *et al.*, 2024).

Later experiments expanded into shuttle missions. During the European Space Agency’s Biorack program in the 1980s, *Drosophila melanogaster* eggs were sent into orbit to study development under microgravity (Table 1). These embryos developed differently compared to those on Earth, demonstrating that gravity is more than just a physical force-it is a biological cue that influences processes like cell division and tissue organization (Vernos *et al.*, 1989).

Table 1. Key Early Spaceflight Experiments with *Drosophila melanogaster*

Year	Species	Purpose	Outcome
1947	<i>Drosophila melanogaster</i>	Radiation exposure	Survived flight; first animals beyond Earth
1980s	<i>D. melanogaster</i> eggs	Microgravity effects on development	Altered embryogenesis and tissue organization

Genetic and Developmental Insights

As molecular biology advanced, insects offered researchers unparalleled insights into genetics in space. *D. melanogaster* is a proven workhorse for genetic studies, and in orbit it revealed how spaceflight alters gene expression and developmental pathways.

For example, space exposure disrupted circadian rhythms, shortened lifespan, and changed the activity of genes regulating stress and metabolism (Ma *et al.*, 2015). More strikingly, long duration spaceflights allowed researchers to study multiple generations of flies. These experiments showed that while fruit flies could reproduce and adapt to microgravity, their gene expression patterns shifted significantly upon return to Earth, reflecting the challenges of readaptation (Ogneva *et al.*, 2016).

Further studies revealed how *D. melanogaster* oocytes, or egg cells, respond to space. After flight, these cells exhibited cytoskeletal changes during the critical early period of adapting to gravity, suggesting that reproductive biology is especially sensitive to altered environments (Ogneva *et al.*, 2022). Such findings raise important questions for human reproductive health during future interplanetary missions.

Even at the genomic level, mutations occurred more frequently in space. Work by Japanese scientists in the 1990s confirmed that fruit flies exposed to cosmic radiation developed higher mutation rates, underscoring the genetic risks astronauts face (Ikenaga et al., 1997).

Neurology and Artificial Gravity

Space doesn't just affect reproduction and genetics-it also impacts the nervous system. Recent studies using fruit flies have provided an unexpected window into how microgravity alters brain function.

In 2022, researchers demonstrated that fruit flies exposed to microgravity showed neurological deficits, including impaired motor control. However, when artificial gravity was applied through a centrifuge during spaceflight, many of these effects were reduced. This suggests that artificial gravity could serve as a powerful countermeasure for protecting human neural health on long missions (Mhatre et al., 2022).

Other work has uncovered the neurological effects of hosting complex ecological interactions in orbit. A 2024 study examined fruit flies and their parasitoid wasps, showing that microgravity not only disrupted host development but also altered host-parasitoid interactions, revealing how ecological dynamics themselves change in space (Chou et al., 2024).

Immunity and Disease Resistance

Microgravity alters not just physical structures but also immune defenses. For astronauts, understanding these changes is crucial because infections in space could become life threatening.

Fruit flies have been indispensable here as well. In 2014, researchers discovered that Toll-mediated immune responses in *D. melanogaster* a system remarkably similar to human innate immunity-were disrupted by spaceflight. The flies became less effective at fighting infection when gravity was absent (Taylor et al., 2014).

These results suggest that astronauts may also experience weakened immune responses, a hypothesis supported by reports of viral reactivation in space crews. Insects, therefore, serve as a simple but powerful model for predicting human vulnerabilities.

Food Security: Edible Insects in Space

While model insects like *D. melanogaster* help unravel the mysteries of biology, edible insects like crickets and mealworms may help feed humanity beyond Earth.

The harsh logistics of space travel mean that packing food for long-term missions is impractical. Instead, researchers are exploring insects as sustainable, high-protein food sources. Insects reproduce rapidly, require little space, and can convert organic waste into biomass. 6 Crickets and mealworms, for example, thrive on plant-based diets and produce far fewer greenhouse gases compared to livestock.

Table 2. Comparison of Edible Insect Characteristics for Space Agriculture

Species	Reproduction Rate	Space Resilience	Nutritional Value
<i>Acheta domesticus</i>	High	Microgravity tolerant	Protein-rich
<i>Tenebrio molitor</i>	Moderate	Microgravity tolerant	Protein & fat-rich

A recent 2025 study examined how these insects tolerate microgravity and whether they can be integrated into space food systems (Table 2). The results were promising: both species showed resilience, indicating that farming insects aboard spacecraft or lunar bases could be feasible (Guidetti *et al.*, 2025).

This approach not only ensures nutrition but also promotes sustainability, closing ecological loops in space habitats by recycling waste into edible protein.

Broader Implications for Human Spaceflight

Taken together, these insect studies highlight an essential truth: the challenges humans face in space-radiation, microgravity, altered immunity, and resource scarcity-are mirrored in the insect world. By observing how these small organisms adapt, we gain insights into our own vulnerabilities and potential survival strategies.

For instance, fruit flies demonstrate that life can reproduce and evolve under space conditions, albeit with genetic and developmental shifts. Their nervous system and immune responses reveal the fragility of biological systems when detached from Earth’s gravity. Meanwhile, edible insects suggest solutions for sustainable living beyond Earth, ensuring that food security does not become an insurmountable barrier.

Insects, then, are not mere passengers in space exploration. They are experimental astronauts in their own right-tiny but vital pioneers that illuminate the biological possibilities and challenges of expanding life into the cosmos.

Conclusion

Insects may be small, but their contributions to space exploration are enormous. From the first fruit flies carried aboard rockets in the 1940s to modern experiments aboard the International Space Station, they have helped us unravel how life copes with microgravity, radiation, and other stresses of spaceflight. Research on *Drosophila melanogaster* has shed light on genetic adaptation, reproductive biology, immune function and neurological resilience, providing parallels directly relevant to human health. Studies on edible species such as crickets and mealworms have shown that insects may also form the backbone of sustainable space agriculture, helping astronauts survive and thrive on long missions.

Most importantly, insect studies remind us that the challenge of expanding into space is not merely technological but biological. Spaceflight alters fundamental life processes, from cellular division to immune defense. Insects-fast-breeding, genetically tractable and remarkably resilient-are the perfect sentinels for detecting these shifts. Their story underscores a profound truth: to secure humanity’s future among the stars, we must first understand how even the smallest organisms adapt beyond Earth. Insects are not just stowaways of science they are pioneers of a cosmic future, guiding us toward life beyond our home planet

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